

```
[root@gnubox~]# iwlist wlan0 scan
wlan0 Scan completed :
    Cell 01 - Address: 00:08:5C:52:E9:83
        ESSID:"alynux"
        Mode:Master
        Channel:11
        Frequency:2.462 GHz (Channel 11)
        Quality=92/100 Signal level=-39 dBm Noise level=-78 dBm
        Encryption key:off
        Bit Rates:1 Mb/s; 2 Mb/s; 5.5 Mb/s; 11 Mb/s; 18 Mb/s
            24 Mb/s; 36 Mb/s; 54 Mb/s; 6 Mb/s; 9 Mb/s
            12 Mb/s; 48 Mb/s
        Extra:tsf=00000000fc021187
```

The above command will list out the various wireless networks available with a number of properties.

Then we can set the IP for the interface card using the *ifconfig* command itself:

```
ifconfig wlan0 192.168.0.5
```

If you are using dynamic addressing, you can obtain the IP address as follows:

```
dhclient wlan0
```

The settings that you've configured with the *ifconfig* tool are available until the system reboot. But it's a waste of time if you need to configure it on every system start. And hence we take the aid of network configuration scripts. On an Ubuntu (or any other Debian-based) system, this file is located at */etc/networks/interfaces*, and contains data similar to the following:

```
auto lo
iface lo inet loopback

iface eth0 inet static
address 164.164.32.101
netmask 255.255.255.240
gateway 164.164.32.97
```

It is necessary to learn this scripting in order to play with your network. The syntax for these are as follows:

1. Add the following lines if you want to configure eth0 as the DHCP:

```
auto eth0
iface eth0 inet dhcp
```

2. Add the following files if you want to configure static IP:

```
auto eth0
iface eth0 inet static
```

```
address <ip_address>
netmask <netmask>
gateway <gateway_ip>
```

3. If it is a wireless network, add the following lines along with the above lines:

```
wireless-essid <network_name>
wireless-key <key>
```

Now, to restart the network daemon, execute the following as the root:

```
/etc/init.d/network restart
```

Spoofing a MAC ID

It is a real hassle for cable Internet customers that they are restricted to using a single machine for Internet access. If you want to plug your laptop in your friend's cable Internet connection, you have to call the service provider to refresh the MAC address.

The MAC address is permanent to the hardware and cannot be changed. Since we operate the hardware via the software abstraction layer, it is quite possible to do some software-level cheating for the network card's MAC ID. We can simply spoof it to some other MAC addresses.

You can obtain the original MAC ID from the *ifconfig* output. Mine is as follows:

```
eth0 Link encap:Ethernet HWaddr 00:1C:23:FB:37:22
```

Now, let's change the last part of the MAC ID from 22 to 23:

```
ifconfig eth0 hw ether 00:1C:23:FB:37:23
```

Now, run *ifconfig* again:

```
[root@gnubox ~]# ifconfig eth0
eth0 Link encap:Ethernet HWaddr 00:1C:23:FB:37:23
    inet addr:192.168.0.2 Bcast:192.168.0.255 Mask:255.255.255.0
    BROADCAST MULTICAST MTU:1500 Metric:1
    RX packets:0 errors:0 dropped:0 overruns:0 frame:0
    TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:1000
    RX bytes:0 (0.0 b) TX bytes:0 (0.0 b)
    Interrupt:17
```

Easy enough? Well, let's consider the following instance: suppose you are in a Wi-Fi campus and the access to the wireless network is restricted using MAC addressing. You can simply look at your friend's laptop MAC ID and spoof it. Yes! You are now free to access the network. Have fun!